

AI Internship Report

Sentiment Analysis of Customer Reviews Using Machine Learning & NLP

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PROGRAM & INSTITUTE
B.Tech CSE, IILM

HOST ORGANIZATION
Codec Technologies

Executive Summary

Project Core & Scope

Designed and built an automated Machine Learning pipeline to analyze unstructured customer review text and classify sentiment polarities.

Fostered end-to-end expertise spanning data acquisition, cleaning, vectorization, classical ML, and neural network classification models.

Key Deliverables & Outcomes

Constructed high-performing Logistic Regression and Artificial Neural Network (ANN) benchmark classifiers.

Achieved robust F1-score evaluation metrics and met AICTE & ICAC industry standards for applied AI internship projects.

Internship Framework



Host Organization

Codec Technologies Pvt. Ltd. — 1-month intensive artificial intelligence internship program (08/08/2026 – 08/09/2026).



Industry Standards

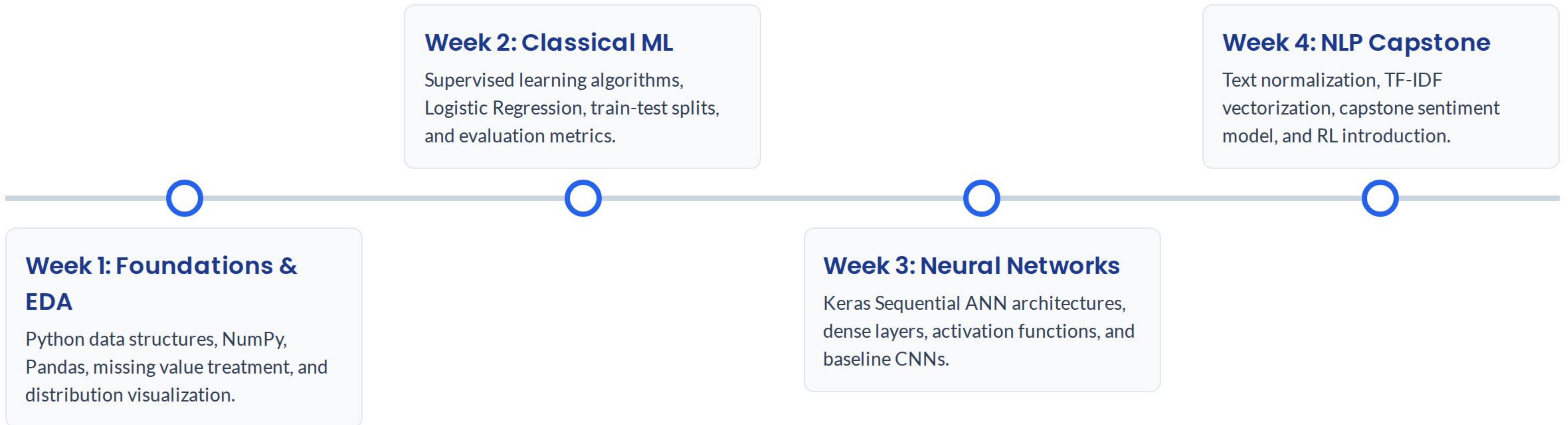
Approved under AICTE & ICAC framework, operating in synchronization with National Internship Portal guidelines.



Academic Context

Fulfills core B.Tech Computer Science Engineering requirements at IILM University, Greater Noida.

4-Week Training Roadmap



Dataset & EDA Insights

10k+

Customer Reviews Processed

Exploratory Data Analysis Highlights

Class Distribution: Addressed sentiment class imbalances between positive, neutral, and negative reviews through stratified sampling.

Text Properties: Analyzed vocabulary density, word length distribution (~45 words/review), and noise factors like HTML tags and punctuation.

Stopword Management: Preserved crucial negation operators (e.g., "not", "no", "never") to prevent sentiment inversion.

NLP Preprocessing Pipeline



1. Text Cleaning

Lowercased review strings, stripped non-alphanumeric noise, URLs, and standardized contractions across records.



2. Tokenization

Segmented text into discrete token streams while selectively filtering uninformative stop-words.



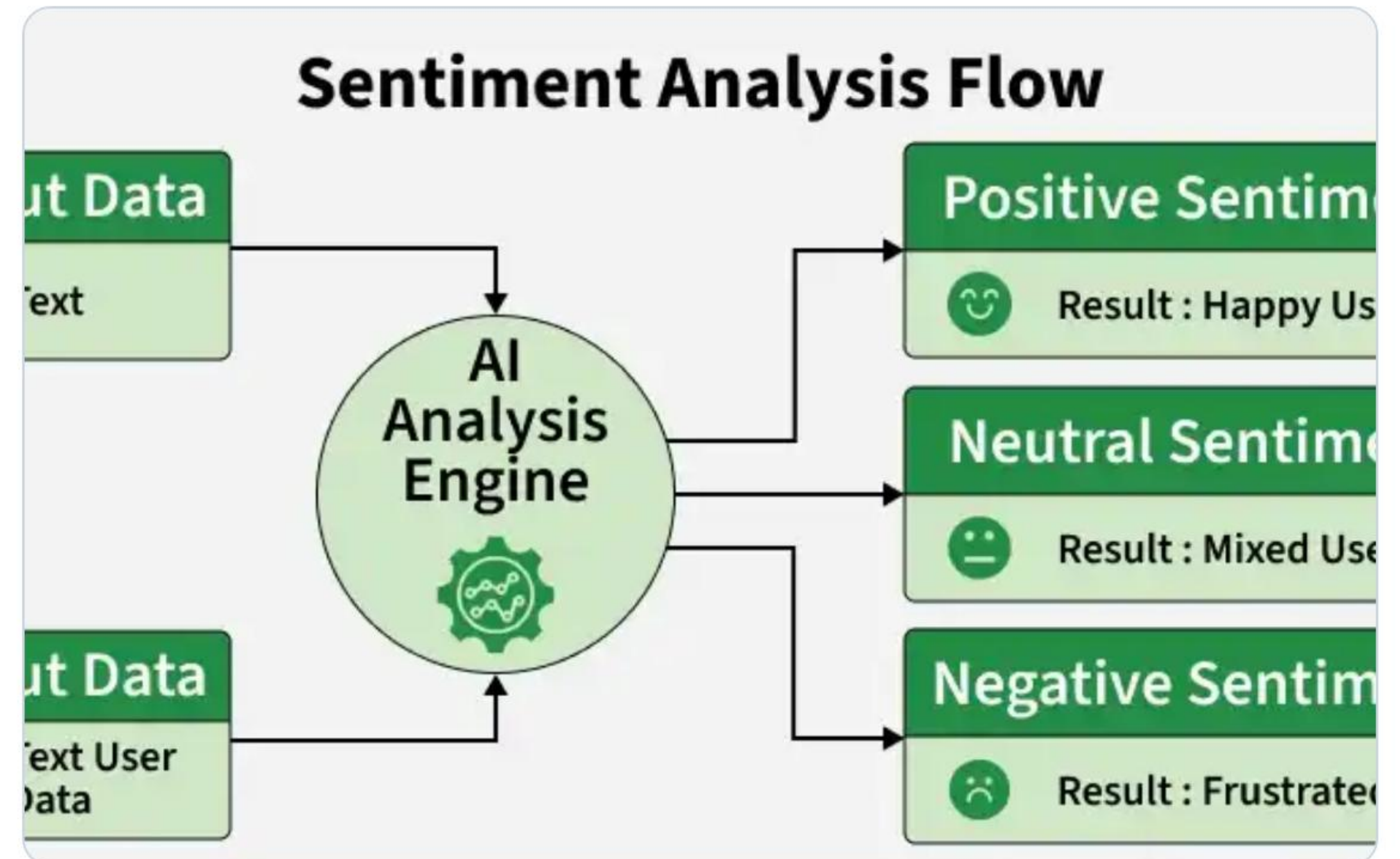
3. TF-IDF Vectorization

Transformed normalized tokens into unigram & bigram sparse TF-IDF feature matrices for model training.

System Architecture

Pipeline Architecture

- **Data Layer:** Raw review ingestion, cleaning, and train-test splits.
- **Feature Layer:** NLTK tokenization and Scikit-Learn TF-IDF vectorizer.
- **Model Layer:** Parallel execution of Logistic Regression and Dense ANN.
- **Evaluation Layer:** Confusion matrix, precision, recall, and F1-score outputs.
- **Tech Stack:** Python, Pandas, Scikit-Learn, TensorFlow/Keras, NLTK.



Model Architectures

Logistic Regression (Baseline)

Serves as a robust linear baseline model optimized using L2 regularization.

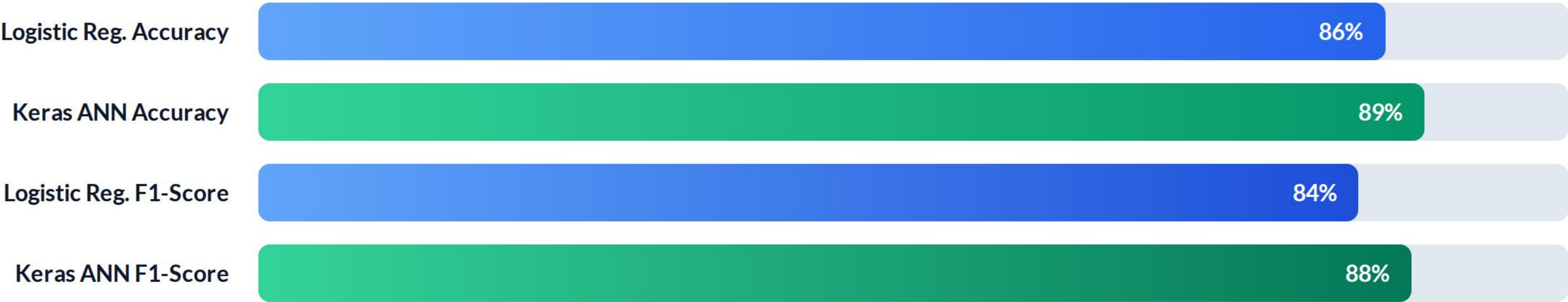
Offers high interpretability, enabling direct inspection of word coefficients driving positive vs. negative predictions.

Keras Sequential ANN

Constructed a Multilayer Perceptron (MLP) featuring dense layers with ReLU activation functions.

Incorporated Dropout regularization (0.3) to prevent overfitting and capture non-linear feature interactions.

Experimental Results



The Keras Sequential ANN achieved an overall performance gain of ~3% in accuracy and 4% in F1-score over Logistic Regression due to dense layer non-linear feature representations.

Evaluation Metrics Criteria

Metric	Target Focus	Mathematical & Conceptual Definition
Accuracy	Overall Correctness	Ratio of correct predictions over total reviews processed.
Precision	False Alarm Reduction	$TP / (TP + FP)$ — Ensures predicted positive sentiment is reliable.
Recall	Sensitivity / Coverage	$TP / (TP + FN)$ — Measures capability to catch all true sentiment instances.
F1-Score	Harmonic Mean	$2 * (Precision * Recall) / (Precision + Recall)$ — Balances skewed labels.
Confusion Matrix	Error Diagnosis	2x2 table detailing True Positives, True Negatives, False Positives & Negatives.

Key Technical Competencies

- ✓ **Core AI Libraries:** Mastered hands-on application of Python, Pandas, Scikit-Learn, TensorFlow/Keras, and NLTK.
- ⚙️ **ML Pipeline Construction:** Developed end-to-end workflows from raw, noisy text ingestion to normalized feature vectors and predictions.
- 📈 **Model Diagnostics:** Applied rigorous evaluation methodologies using confusion matrices, precision-recall trade-offs, and loss curves.
- 💡 **Real-World Problem Solving:** Solved common NLP challenges including text noise, class distribution imbalances, and overfitting.

Conclusion & Questions

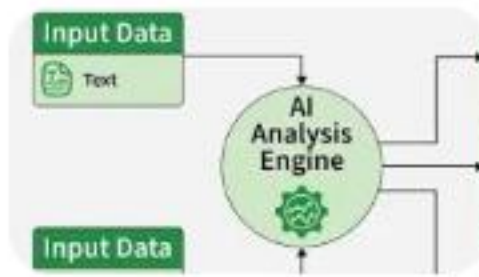
Thank you for your guidance and support throughout the internship!

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Codec Technologies Pvt. Ltd. Internship Project

Image Sources



https://media.geeksforgeeks.org/wp-content/uploads/20260529113730211990/sentiment_analysis_flow.webp

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